Circuit Analysis (Lab # 2)

Experiment 2: Measurements of Waveforms and Fundamental Quantities, Basic resistive circuits.

Name: Talha Yousaf **Registration No: 2025-CE-93**

Date: 16-3-2025 **Grade and Signature: ……**

CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	**Level1 (1)**	**Level2 (2-3)**	**Level3 (4-5)**	**Level4 (6-7)**
Report Marks (3)			**Total marks (10)**	

Objectives: This experiment will provide exposure to measurements of waveforms and Fundamental Quantities, Basic resistive circuits. A primary purpose of this lab course is for you to master the use of electronic test equipment and measurement of fundamental quantities.

Equipment required:

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter.

Procedure

Experiment # 1

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'A' shown below in Figure 1.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 100 Hz and peak-to peak voltage of 5V.
4. Set up one probe across input signal, and another across R1 and R2 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

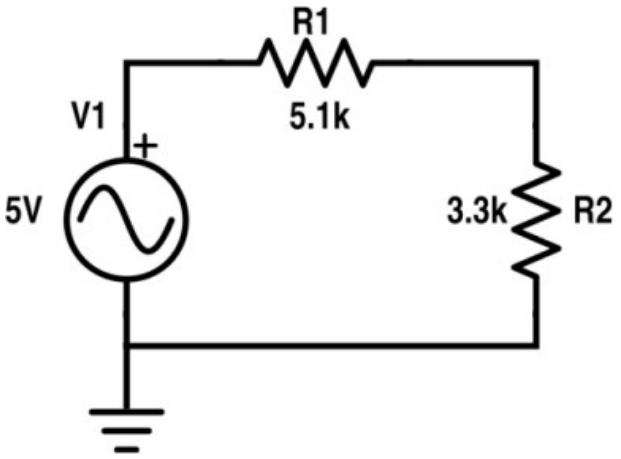

Figure 1 Circuit 'A'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	5V	1.08	3.04V
R2	5V	0.69	1.96V

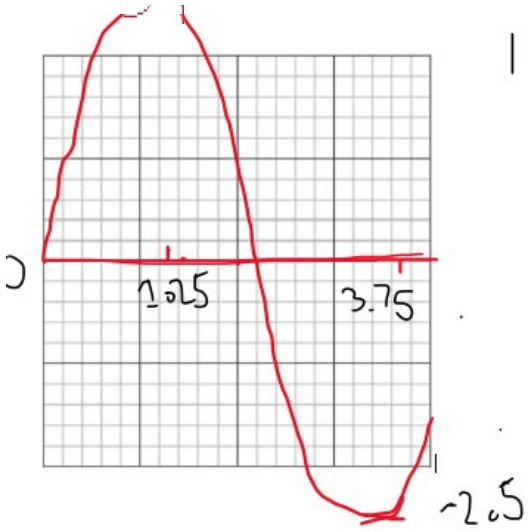

Figure 2 Input Voltage Signal

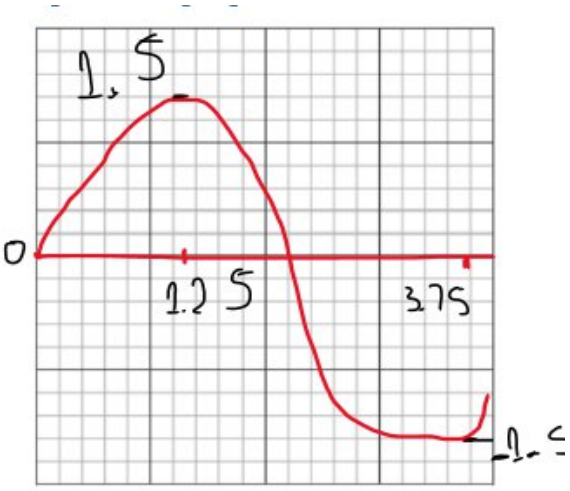

Figure 3 Voltage Signal across R1

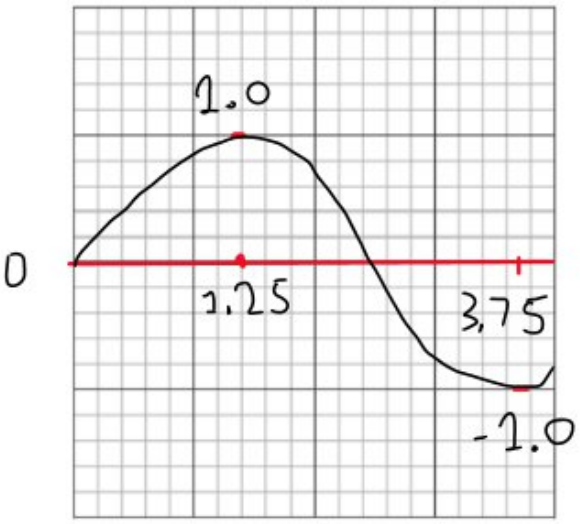

Figure 5 Voltage Signal across R2

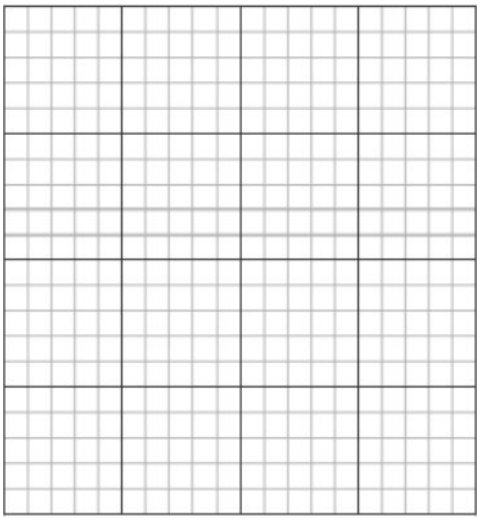

Figure 6 Draw 7V Sinusoidal

Experiment # 2

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'B' shown below in Figure 7.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 200 Hz and peak-to peak voltage of 7V.
4. Set up one probe across input signal, and another across R1, R2 and R3 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

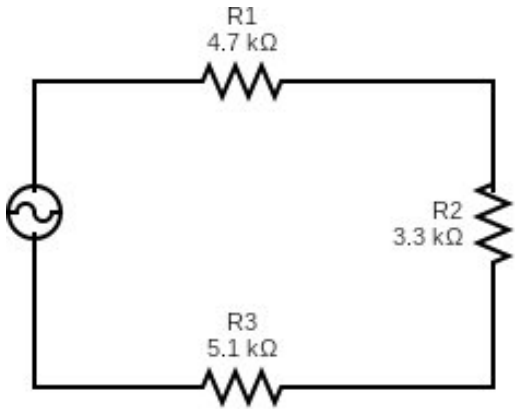

Figure 7 Circuit 'B'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	7V	0.87V	2.51V
R2	7V	0.62V	1.76V
R3	7V	0.95V	2.73V
R1 and R2	7V	1.51V	4.27V
R2 and R3	7V	1.58V	4.49V

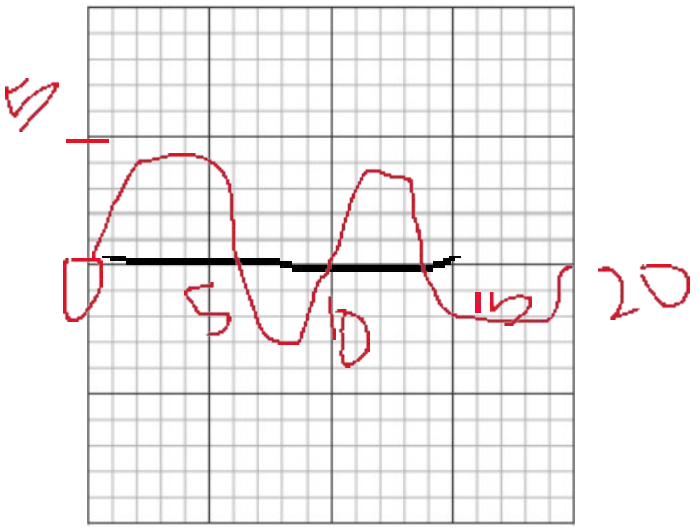

Figure 8 Input Voltage Signal

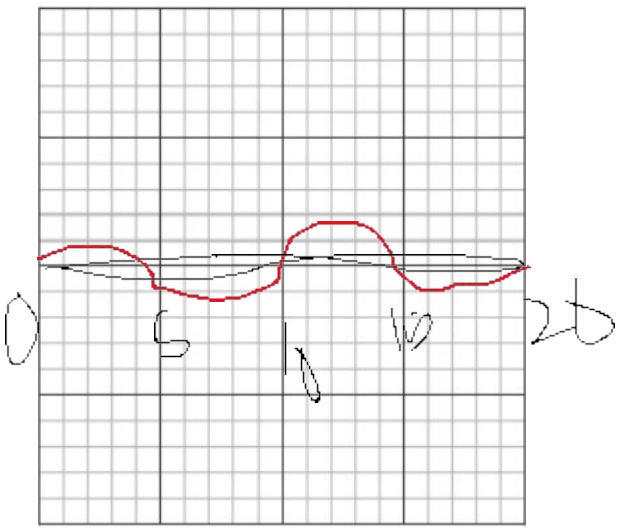

Figure 9 Voltage Signal across R1

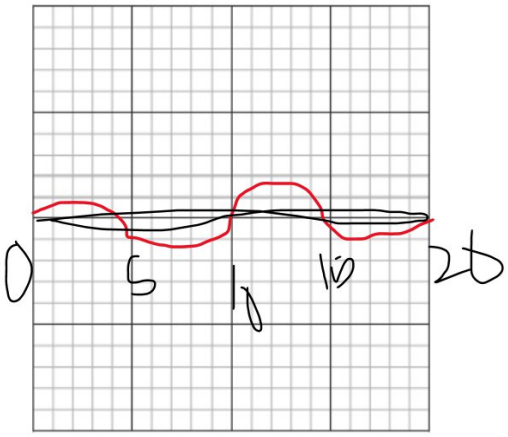

Figure 10 Voltage Signal across R2

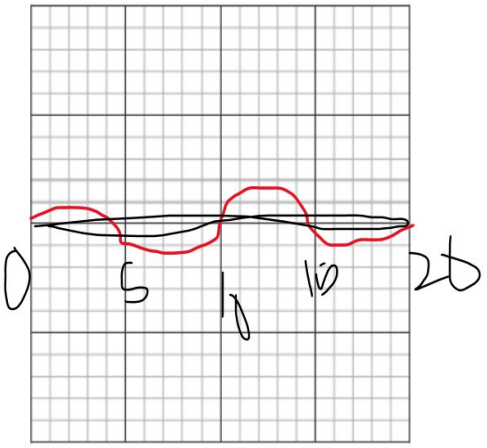

Figure 11 Voltage Signal across R3

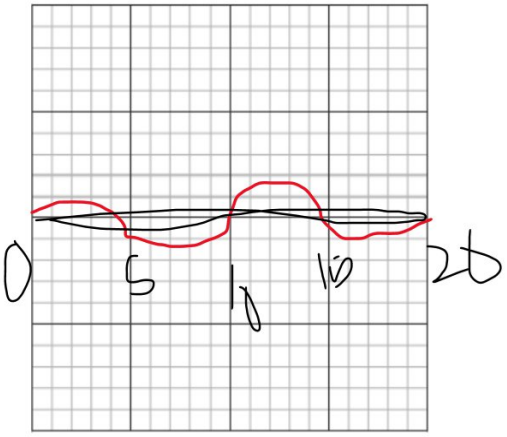

Figure 12 Voltage Signal across R1 and R2

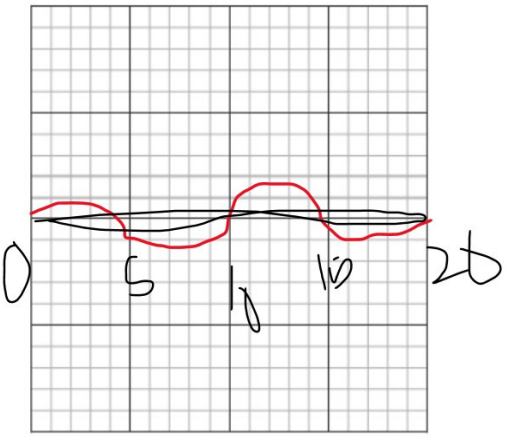

Figure 13 Voltage Signal across R2 and R3

Post Lab Questions

1. Define Ohm's law.

Ohm's Law **states that the** current flowing through a conductor is directly proportional to the voltage across it

V=IR

2. What is resistor?

A resistor is a passive electronic component used to limit or control the flow of electric current in a circuit**.**

It provides resistance and protects other components from excessive current**.**

3. How resistor works?

A resistor works by opposing the flow of electric current.

When current flows through the resistor:

- The **resistor** reduces the current
- Some electrical energy is converted into heat
- It helps control voltage and current levels in the circuit

4. What is your inference from the input and output waveforms on oscilloscope?

From the oscilloscope waveform we can observe:

- The **shape of the signal** (sine, square, triangular)
- The **amplitude of the signal**
- The **frequency and time period**
- Any **phase shift or distortion**

Show calculations of 7V input signal (Figure10) from signal generator.

a. Voltage calculations(Following volt/ dev and number of boxes of oscilloscope)
b. Frequency/Period calculations(Following Steps in frequency measurements section)
c. Current calculation(Ohm Law)

Part A

Voltage=(Volt/Div)×(Number of Vertical Divisions)

Volt/Div = **2 V**
Number of boxes = **3.5 divisions**

Voltage=2×3.5

So the **input signal voltage = 7 V**.

Part B

Time/Div = **1 ms**
Boxes for one cycle = **4 divisions**

T=(Time/Div)×(Number of divisions)

$T=1ms\times4$
$T=4ms$

Now frequency:

$f=1/T$
$f= \{1\}/\{0.004\}$
$f=250$ Hz

Part C

$I=VR$

Example:

Voltage = **7 V**
Resistance = **1 kΩ = 1000 Ω**

$I= \{7\}/\{1000\}$
$I=0.007$ A

So the **current in the circuit = 7 mA**